

A disproof of the conjectured generating function and recurrence for OEIS A141135

Adrian Perez Fontelles

Independent researcher

31 August 2026

Abstract

OEIS A141135 carries two conjectures contributed together by Colin Barker: a rational generating function and a linear recurrence of order four. Both are false, and both are refuted by the entry's own published terms. The generating function reproduces $a(1), \dots, a(23)$ and then disagrees at $n = 24, 27$ and 30 ; the recurrence holds from its stated starting point until $n = 24$, where it fails, and fails again at $n = 25$. The argument is a finite exact computation and is reproduced in full below, so the reader can check it by hand. We also record why a disagreement of this shape is a refutation of the conjecture rather than a doubt about the data.

2020 Mathematics Subject Classification. 05A15, 11B37, 52C20.

1 The sequence and the conjectures

OEIS A141135 is “Minimal number of unit edges required to construct n regular pentagons when allowing edge-sharing”. It has offset 1 and its published DATA field carries the thirty terms

5, 9, 13, 17, 21, 24, 28, 32, 36, 39,
43, 47, 50, 54, 58, 61, 65, 69, 72, 76,
80, 83, 87, 90, 94, 98, 101, 105, 109, 112.

Each $a(n)$ is the minimum of a finite set — the edge counts of the finitely many edge-sharing configurations of n unit-edge regular pentagons — so each term is a definite integer, computed and published, not an estimate.

The entry carries the following:

Conjectures from *Colin Barker*, Apr 05 2019: (Start)

G.f.: $x(5 + 4x + 4x^2 - x^3 - x^5 + x^8 - x^9)/((1 - x)^2(1 + x + x^2))$.

$a(n) = a(n - 1) + a(n - 3) - a(n - 4)$ for $n > 10$.

(End)

As of the “Last modified” line on the live entry (2025-01-05, revision 24) both are still recorded as conjectures.

Write

$$G(x) = \frac{x(5 + 4x + 4x^2 - x^3 - x^5 + x^8 - x^9)}{(1 - x)^2(1 + x + x^2)}, \quad G(x) = \sum_{n \geq 1} g(n) x^n, \quad (1)$$

and let

$$R(n) = a(n) - a(n - 1) - a(n - 3) + a(n - 4). \quad (2)$$

The two conjectures assert $g(n) = a(n)$ for all $n \geq 1$ and $R(n) = 0$ for all $n > 10$.

2 The generating function

Proposition 1. $g(n) = a(n)$ for $1 \leq n \leq 23$, while

$$g(24) = 91, \quad g(27) = 102, \quad g(30) = 113,$$

against the published values $a(24) = 90$, $a(27) = 101$, $a(30) = 112$.

Proof. The denominator of (1) is $(1-x)^2(1+x+x^2) = 1-x-x^3+x^4$, whose constant term is 1, so the coefficients $g(n)$ are obtained from the numerator by exact long division: writing the numerator as $\sum_k u_k x^k$ and the denominator as $\sum_k v_k x^k$ with $v_0 = 1$,

$$g(n) = u_n - \sum_{k \geq 1} v_k g(n-k) = u_n + g(n-1) + g(n-3) - g(n-4).$$

Every step is an integer subtraction. Carrying it out gives

$$\begin{aligned} g(1), \dots, g(30) = & 5, 9, 13, 17, 21, 24, 28, 32, 36, 39, \\ & 43, 47, 50, 54, 58, 61, 65, 69, 72, 76, \\ & 80, 83, 87, \mathbf{91}, 94, 98, \mathbf{102}, 105, 109, \mathbf{113}, \end{aligned}$$

which agrees with the published terms except in the three positions shown in bold. $\square$

Theorem 2. *The conjectured generating function of A141135 is incorrect.*

Proof. By Proposition 1, $g(24) = 91 \neq 90 = a(24)$. A generating function whose coefficient at x^{24} differs from $a(24)$ is not the generating function of a . $\square$

3 The recurrence

Theorem 3. *The conjectured recurrence $a(n) = a(n-1) + a(n-3) - a(n-4)$ fails at $n = 24$ and at $n = 25$, both of which lie in the range $n > 10$ where it is asserted.*

Proof. From the published terms,

$$a(23) = 87, \quad a(21) = 80, \quad a(20) = 76, \quad a(24) = 90,$$

so

$$a(23) + a(21) - a(20) = 87 + 80 - 76 = 91 \neq 90 = a(24),$$

that is $R(24) = 90 - 87 - 80 + 76 = -1$. Likewise with $a(24) = 90$, $a(22) = 83$, $a(21) = 80$ and $a(25) = 94$,

$$a(24) + a(22) - a(21) = 90 + 83 - 80 = 93 \neq 94 = a(25),$$

that is $R(25) = +1$. Evaluating $R(n)$ at every n from 11 to 30 gives zero except at these two indices. $\square$

Remark 4. The two conjectures are equivalent statements, so it is no accident that both fail. The denominator $1 - x - x^3 + x^4$ of (1) is exactly the characteristic polynomial of (2), so a rational generating function with that denominator and a numerator of degree 10 is the same assertion as the recurrence holding for $n > 10$. What Proposition 1 adds is the location of the first failure, $n = 24$.

4 Why this refutes the conjecture rather than the data

A disagreement between a conjectured formula and a published sequence could in principle be resolved either way. Here it cannot.

The terms of A141135 are minima over finite sets of configurations: for each n there are finitely many ways to place n edge-sharing unit-edge regular pentagons, each uses a definite number of unit edges, and $a(n)$ is the least such number. A published term is therefore the outcome of a computation, and the entry's terms have stood since the sequence was contributed.

The conjectured generating function, by contrast, was obtained by fitting: a rational function with a degree-10 numerator and a degree-4 denominator has 15 free coefficients, and the entry's first 23 terms determine such a fit comfortably. That the fit then breaks at the twenty-fourth term is the expected failure mode of fitting, and it is what the computation above exhibits.

The failures are also not isolated noise. They occur at $n = 24, 27, 30$ — a step of three, the period of the factor $1 + x + x^2$ in the denominator — so the fitted function drifts from the sequence by one at every third index from 24 onward, and the published range simply stops at 30.

5 Verification

The long division of Proposition 1 was carried out in exact integer arithmetic, and independently the residual $R(n)$ of (2) was evaluated on the published terms at every index from 11 to 30. The two computations agree: $R(n) = 0$ except $R(24) = -1$ and $R(25) = +1$, and $g(n) - a(n) = 0$ except at $n = 24, 27, 30$ where it is $+1$. No floating-point arithmetic was used at any point, and no term was taken from anywhere but the entry's own DATA field.

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry [A141135](#), 2025.